

8086 Shift Instructions

1. Overview of Shift Instructions

Concept

Shift instructions move bits **left or right** within an operand. They are used for:

- Multiplication or division by powers of 2
- Bit extraction and masking
- Arithmetic sign handling

Rules / Notes

General Syntax

```
SHL destination, count
SHR destination, count
SAL destination, count
SAR destination, count
```

Count can be:

- Immediate value = 1
- CL register

2. SHL / SAL (Shift Left / Shift Arithmetic Left)

Concept

SHL and **SAL** are identical in 8086. They shift all bits to the left, inserting 0 into the least significant bit.

Rules / Notes

Operation

LSB $\leftarrow$ 0

Effect

- Equivalent to multiplication by 2^n
- MSB shifted out goes into CF

Example

Bit-Level Example

0010 1101 $\xrightarrow{SHL}$ 0101 1010

Example

```
MOV AL, 2DH      ; 0010 1101
SHL AL, 1
; AL = 5AH
```

Example

```
MOV AX, 0003H
SHL AX, 2          ; AX = 000CH (3 x 4)
```

Exam Tip

SHL/SAL is commonly used to multiply numbers by powers of 2.

3. SHR (Shift Logical Right)

Concept

SHR shifts bits to the right. Zeros are inserted into the most significant bit.

Rules / Notes

Operation

MSB $\leftarrow$ 0

Effect

- Equivalent to **unsigned division** by 2^n
- LSB shifted out goes into CF

Example

Bit-Level Example

1010 1100 $\xrightarrow{SHR}$ 0101 0110

Example

```
MOV AL, ACH
SHR AL, 1
; AL = 56H
```

Example

```
MOV AX, 0010H
SHR AX, 2      ; AX = 0004H
```

Exam Tip

Use SHR only for **unsigned division**. Signed numbers lose their sign.

4. SAR (Shift Arithmetic Right)

Concept

SAR shifts bits to the right while preserving the sign bit. The MSB is filled with the original sign bit.

Rules / Notes

Operation

$\text{MSB} \leftarrow \text{Original MSB}$

Effect

- Signed division by 2^n
- Preserves sign of the number

Example

Bit-Level Example (Negative Number)

$1111\ 0100 \xrightarrow{\text{SAR}} 1111\ 1010$

Example

```
MOV AL, F4H      ; -12 (signed)
SAR AL, 1
; AL = FAH (-6)
```

Example

```
MOV AX, FF9CH    ; -100
SAR AX, 2
```

Exam Tip

Use SAR for **signed division**. SHR would destroy the sign bit.

5. Flags Affected by Shift Instructions

Rules / Notes

- **CF**: last bit shifted out
- **ZF**: set if result is zero
- **SF**: reflects sign of result
- **PF**: parity of result
- **OF**: defined only for 1-bit shifts

Exam Tip

For multi-bit shifts, **OF is undefined**. Exam questions often test this detail.

Quick Revision (One Look)

- **SHL / SAL**: left shift, multiply by 2^n
- **SHR**: logical right shift, unsigned divide by 2^n
- **SAR**: arithmetic right shift, signed divide by 2^n
- Shift count = 1 or CL
- CF stores shifted-out bit